Scale Bridge and Zone Node Wiring Diagrams

Oct 6, 2026 · @Kavindu Jayasundara

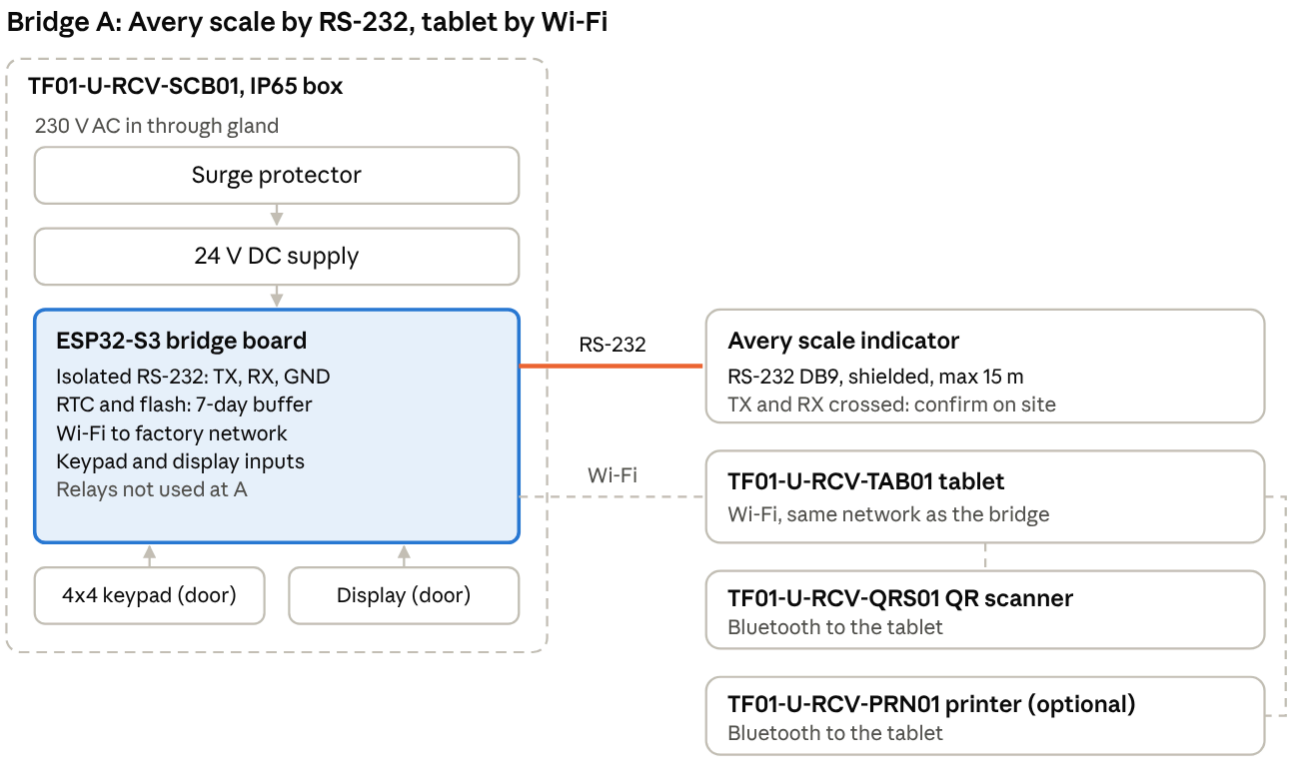
Key and conventions

This doc covers 9 build variants: 4 scale bridge types (7 units) and 5 zone node types (18 units). Each diagram shows the enclosure on the left as a dashed outline and the field devices on the right, with every run labelled with its signal and cable.

- **Device IDs:** SITE-FLOOR-ZONE-TYPEnn, as in the v2 design (TF01-U-TR03-THS02 is trough 3, upper floor, temperature/humidity sensor 2). Each diagram shows the ID of one example unit, and the table gives the IDs for the other units of that type.
- **RS-485 bus:** daisy-chained, never star-wired, with a 120 Ω terminator at the far end only. The node end has built-in termination; confirm on site.
- **Bus cable:** 2-pair shielded twisted pair. Pair 1 carries A and B, pair 2 carries +24 V and 0 V for the sensors. Ground the shield at the node end only.
- **Bus settings (proposed):** 9600 baud, 8N1. Confirm against each sensor's datasheet.
- **Modbus addresses (proposed, installer confirms):** temperature/humidity 1–9, current modules 21–29, IR sensors 31–38, thermocouple module 41. Addresses only need to be unique on each node's own bus.
- **Terminal names** (+24 V, 0 V, RS-485 A/B, RS-232 TX/RX/GND, relay R1–R3) are functional labels. Map them to the terminal numbers of the controller you buy, and mark any unknown detail as "confirm on site".

Scale bridge A: receiving

One unit, TF01-U-RCV-SCB01. The bridge reads the scale. The tablet handles supplier, area, maturity and photos, and talks to the bridge over Wi-Fi.



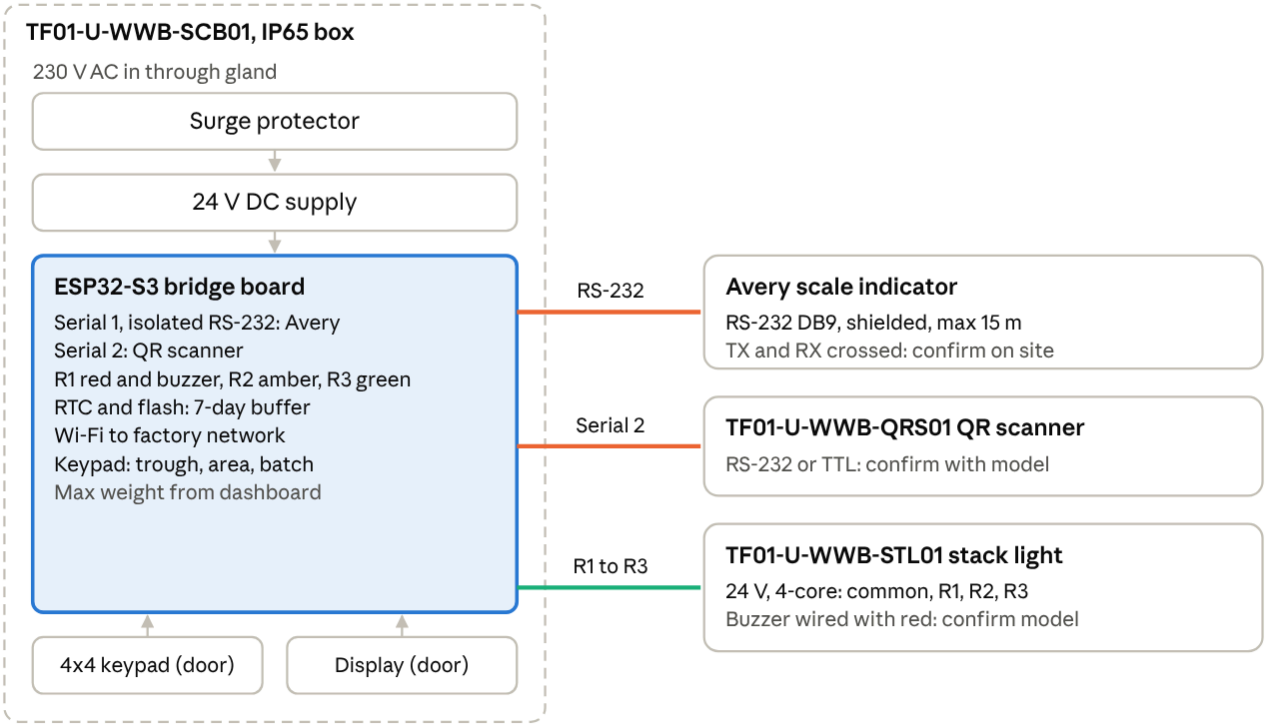
scale bridge A · receiving

Bridge terminal	Connects to	Device ID	Cable
RS-232 TX, RX, GND	Avery indicator data port	Scale (no ID)	Shielded RS-232, DB9, max 15 m
Wi-Fi	Rugged tablet	TF01-U-RCV-TAB01	None
Tablet Bluetooth	QR scanner	TF01-U-RCV-QRS01	None
Tablet Bluetooth	Receipt printer, optional	TF01-U-RCV-PRN01	None
+24 V, 0 V	24 V DC supply	None	Internal
Keypad, display ports	Door-mounted keypad and display	None	Internal ribbon

Scale bridge B: withered weigh

One unit, TF01-U-WWB-SCB01. The relays switch the stack light locally, so the over-limit alarm works without a network.

Bridge B: scale and scanner by serial, stack light by relays

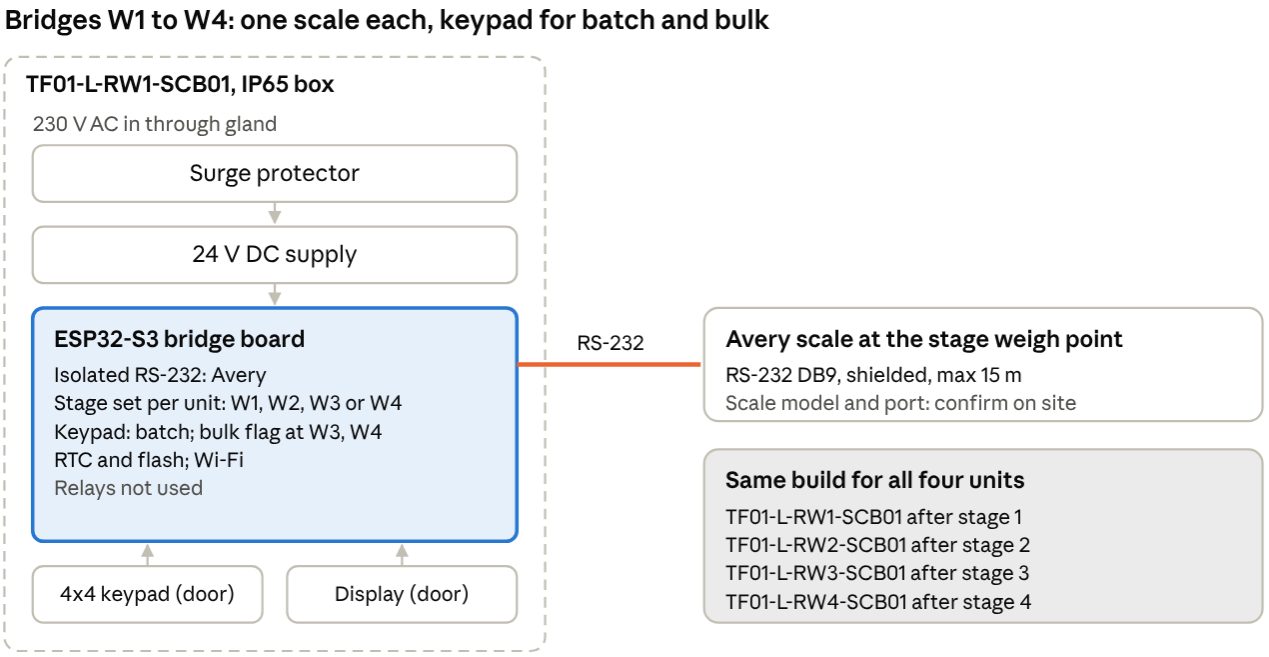


scale bridge B · withered weigh

Bridge terminal	Connects to	Device ID	Cable
Serial 1, RS-232 TX, RX, GND	Avery indicator data port	Scale (no ID)	Shielded RS-232, DB9, max 15 m
Serial 2	QR scanner	TF01-U-WWB-QRS01	Scanner cable, RS-232 or TTL (confirm on site)
R1	Stack light red and buzzer	TF01-U-WWB-STL01	4-core, 24 V
R2	Stack light amber	TF01-U-WWB-STL01	Same cable
R3	Stack light green	TF01-U-WWB-STL01	Same cable
Relay common	+24 V	None	Same cable
+24 V, 0 V	24 V DC supply	None	Internal
Keypad, display ports	Door-mounted keypad and display	None	Internal ribbon

Scale bridges W1 to W4: rolling

Four identical units, one per stage weigh point. Each unit's stage is set in its configuration, so the operator only enters the batch and, at W3 or W4, the bulk flag.



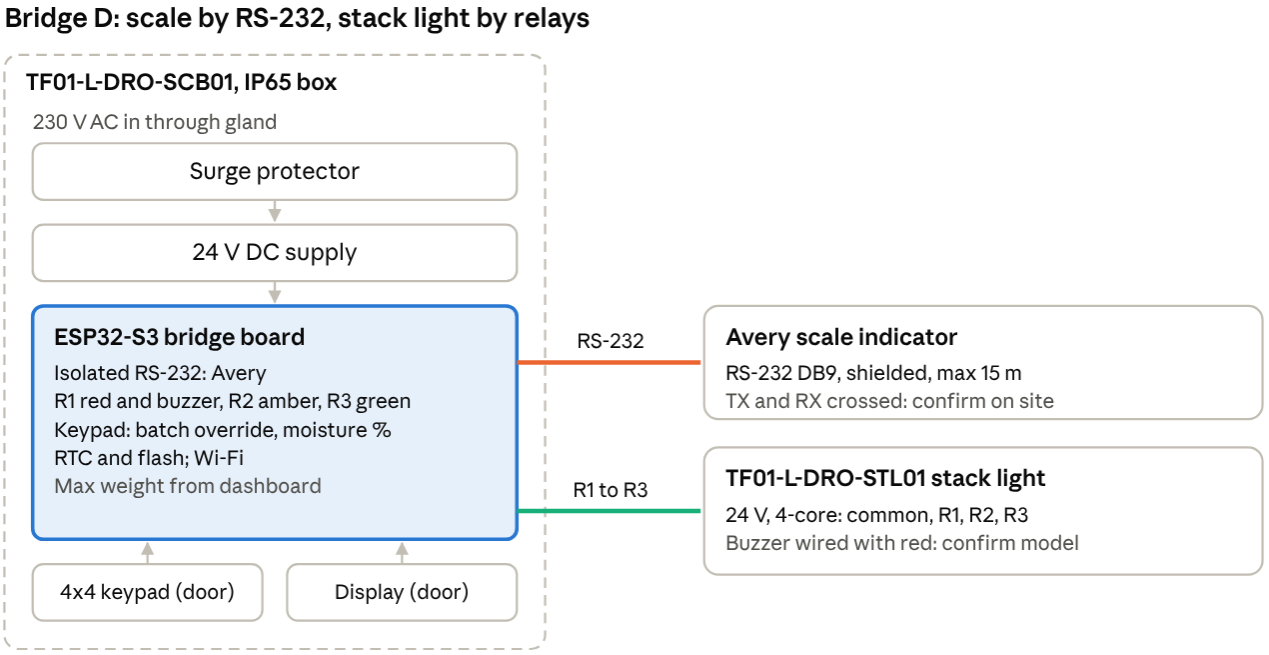
scale bridges W1 to W4 · rolling

Bridge terminal	Connects to	Device ID	Cable
RS-232 TX, RX, GND	Avery indicator at that stage	Scale (no ID)	Shielded RS-232, DB9, max 15 m
+24 V, 0 V	24 V DC supply	None	Internal
Keypad, display ports	Door-mounted keypad and display	None	Internal ribbon

Units: TF01-L-RW1-SCB01, TF01-L-RW2-SCB01, TF01-L-RW3-SCB01, TF01-L-RW4-SCB01.

Scale bridge D: drier output

One unit, TF01-L-DRO-SCB01. Same as B without the scanner. The keypad also takes the made-tea moisture reading from the handheld meter.



scale bridge D · drier output

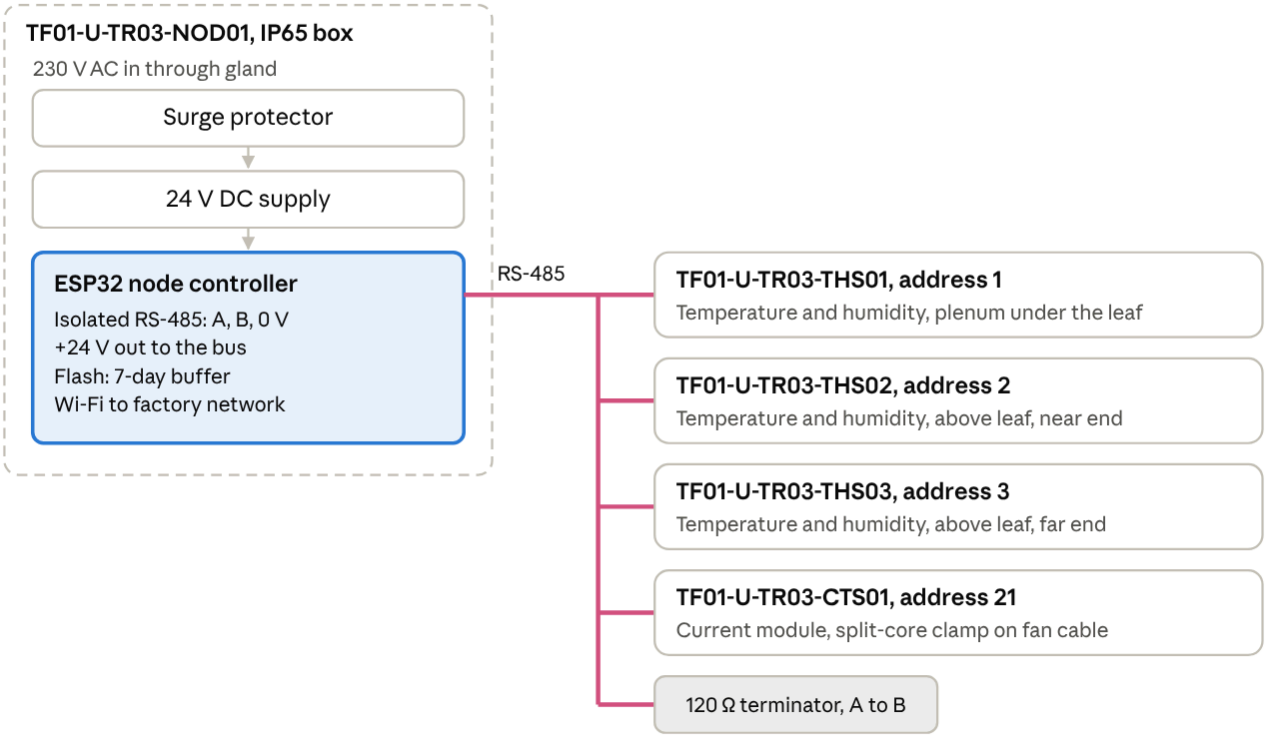
Bridge terminal	Connects to	Device ID	Cable
RS-232 TX, RX, GND	Avery indicator data port	Scale (no ID)	Shielded RS-232, DB9, max 15 m
R1	Stack light red and buzzer	TF01-L-DRO-STL01	4-core, 24 V
R2	Stack light amber	TF01-L-DRO-STL01	Same cable
R3	Stack light green	TF01-L-DRO-STL01	Same cable
Relay common	+24 V	None	Same cable
+24 V, 0 V	24 V DC supply	None	Internal
Keypad, display ports	Door-mounted keypad and display	None	Internal ribbon

Trough node, 11 units

Each trough has one node with the same build. Only the trough number in the IDs changes: TF01-U-TR01-NOD01 to TF01-U-TR11-NOD01. Mount the node near the fan end, so the bus runs from the plenum sensor to the far end.

Trough node: four devices daisy-chained on one RS-485 bus

Bus: 2-pair shielded twisted pair (A, B, +24 V, 0 V), wired in the order shown. Example: trough 3.



trough node · 4 Modbus devices, 1 bus

Order on bus	Device ID (trough nn)	Device	Modbus address (proposed)	Cable
1	TF01-U-TRnn-THS01	Temperature/humidity, plenum	1	2-pair STP
2	TF01-U-TRnn-THS02	Temperature/humidity, above leaf, near end	2	2-pair STP
3	TF01-U-TRnn-THS03	Temperature/humidity, above leaf, far end	3	2-pair STP
4	TF01-U-TRnn-CTS01	Current module, clamp on the 7.5 hp fan supply	21	2-pair STP; clamp lead as supplied
End	Terminator	120 Ω across A and B	None	None

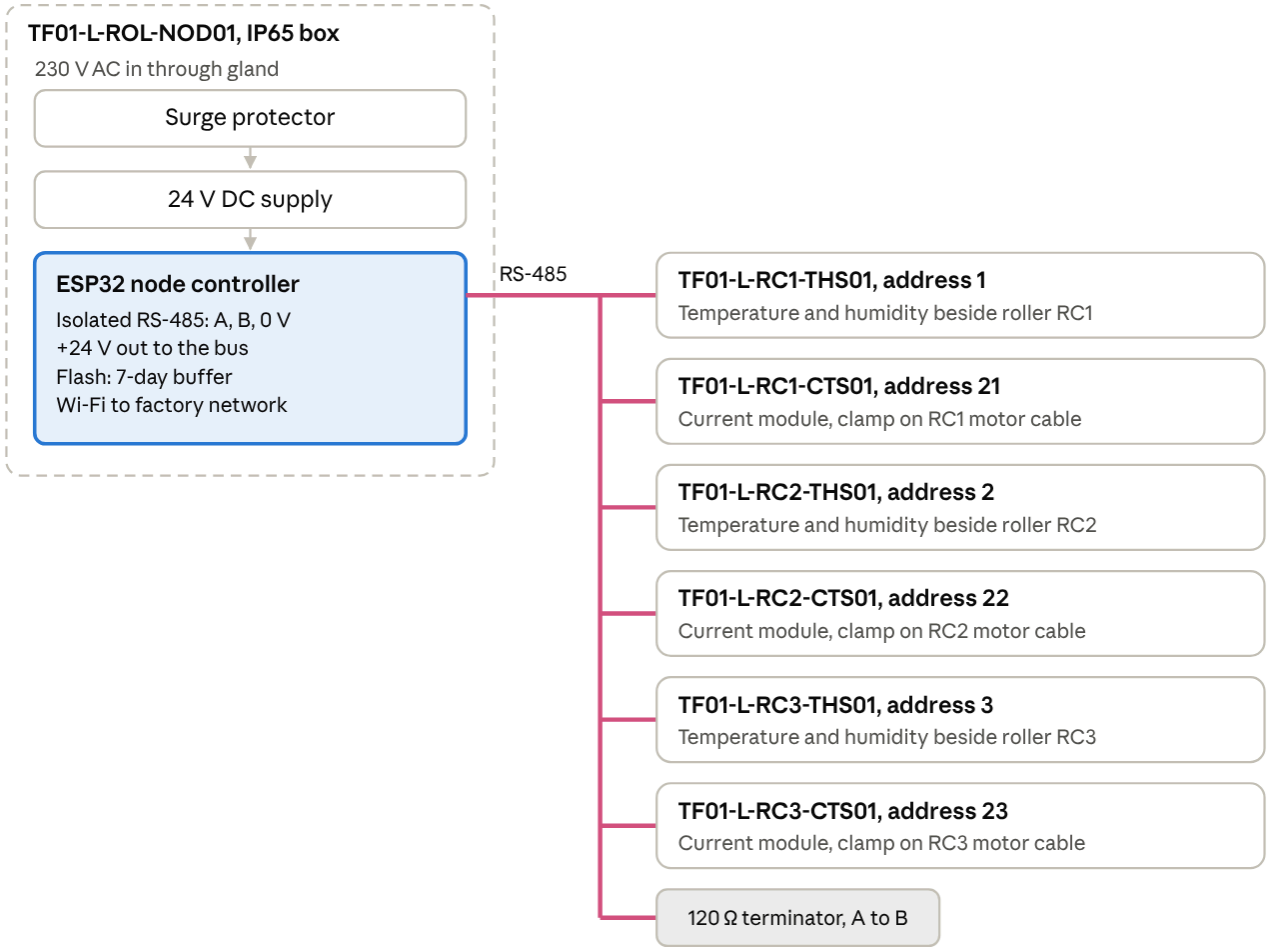
The current clamp goes around one phase conductor of the fan supply, not the whole cable. A qualified electrician fits it.

Rolling nodes, 3 units

Each rolling node serves three rollers, with one temperature/humidity sensor and one motor current module per roller. Wire them roller by roller along the floor.

Rolling node: three rollers, two devices each, one bus

NOD01 shown (charging rollers). NOD02 and NOD03 are wired the same way; see the table.



rolling node NOD01 · 6 Modbus devices, 1 bus

Node	Order on bus	Device ID	Modbus address (proposed)
TF01-L-ROL-NOD01	1 to 6	RC1-THS01, RC1-CTS01, RC2-THS01, RC2-CTS01, RC3-THS01, RC3-CTS01	1, 21, 2, 22, 3, 23
TF01-L-ROL-NOD02	1 to 6	R21-THS01, R21-CTS01, R22-THS01, R22-CTS01, R23-THS01, R23-CTS01	1, 21, 2, 22, 3, 23
TF01-L-ROL-NOD03	1 to 6	R31-THS01, R31-CTS01, R32-THS01, R32-CTS01, R41-THS01, R41-CTS01	1, 21, 2, 22, 3, 23

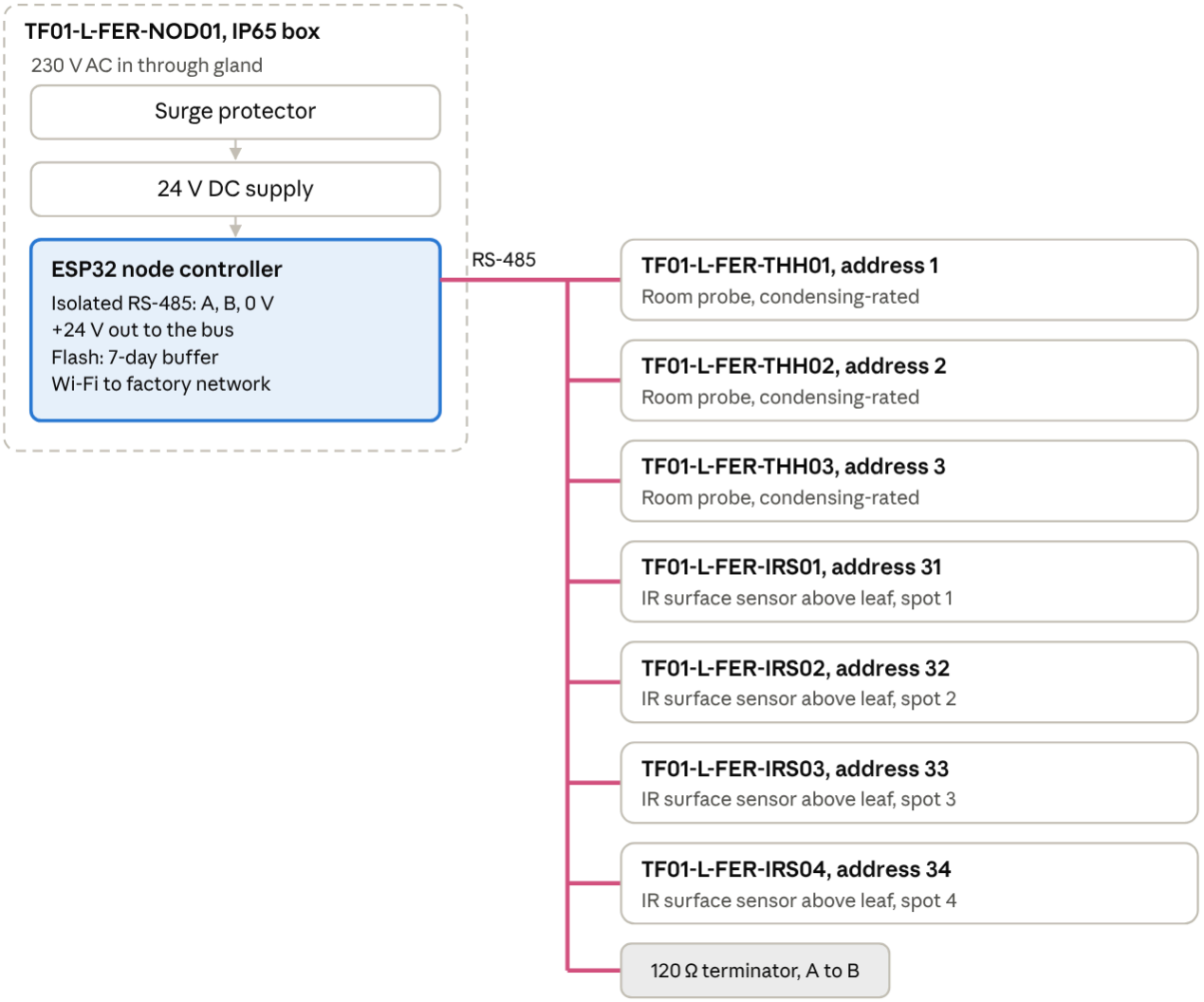
All IDs carry the TF01-L- prefix. Cable is 2-pair STP throughout, with a 120 Ω terminator after the last device on each bus.

Fermentation nodes, 2 units

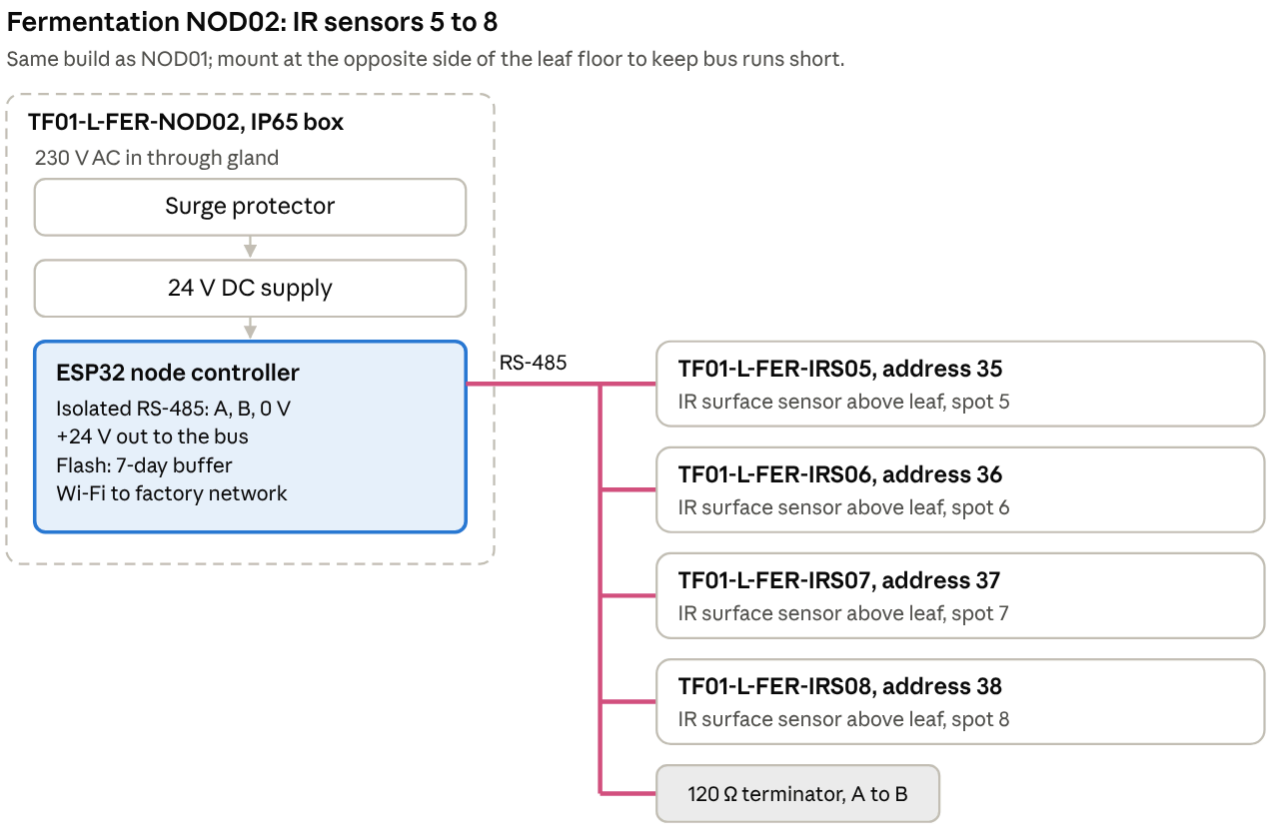
The 8 IR sensors are split across two nodes so no bus run crosses the whole room. These diagrams assume Modbus IR sensors. If a 4–20 mA model is bought instead, wire it to the node's analogue inputs (confirm on site).

Fermentation NOD01: three room probes and IR sensors 1 to 4

Bus: 2-pair shielded twisted pair, wired in the order shown. Box must stay sealed: room air is near 100% RH.



fermentation NOD01 · 7 Modbus devices



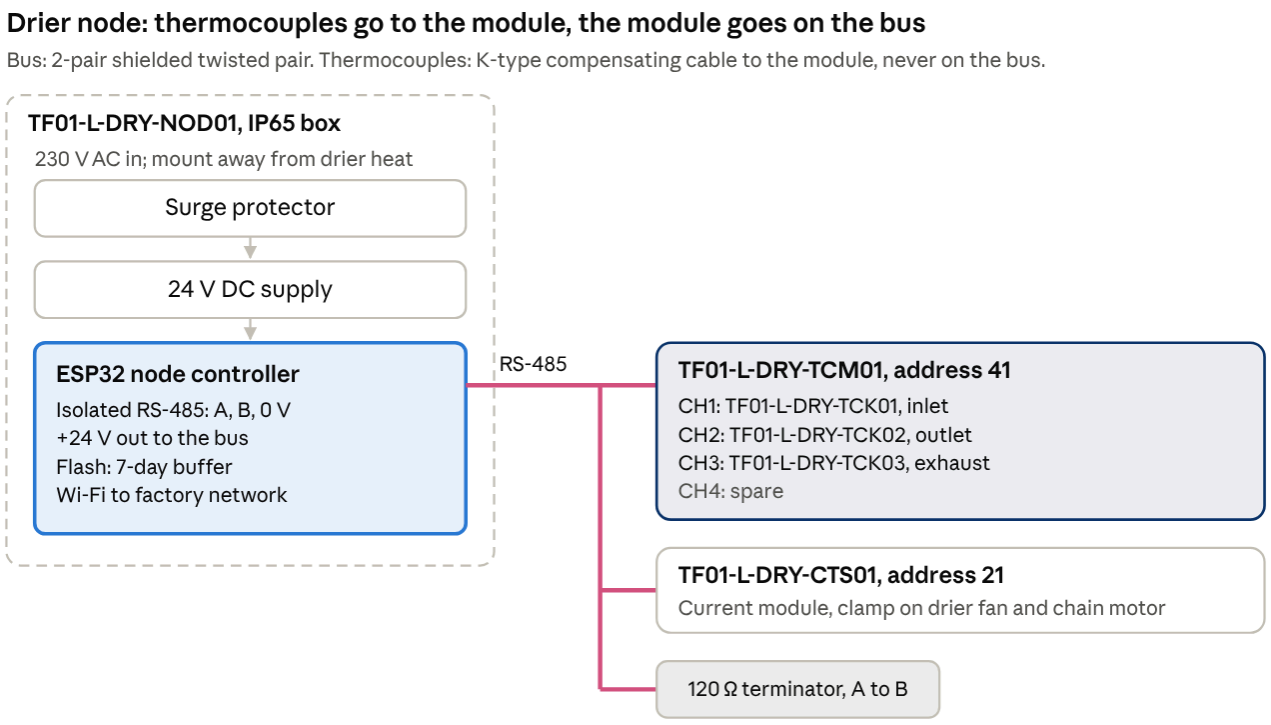
fermentation NOD02 · 4 Modbus devices

Node	Order on bus	Device ID	Device	Modbus address (proposed)
TF01-L-FER-NOD01	1 to 3	TF01-L-FER-THH01 to THH03	Room temperature/humidity, condensing-rated	1, 2, 3
TF01-L-FER-NOD01	4 to 7	TF01-L-FER-IRS01 to IRS04	IR surface sensor, spots 1 to 4	31 to 34
TF01-L-FER-NOD02	1 to 4	TF01-L-FER-IRS05 to IRS08	IR surface sensor, spots 5 to 8	35 to 38

Aim each IR sensor straight down at its spot, at a height that keeps its view on the leaf (check the sensor's field-of-view on site). Mount the room probes away from doors and humidifier outlets.

Drier node, 1 unit

The three thermocouples connect to a 4-channel thermocouple module, and only the module sits on the RS-485 bus. Keep the node box out of the drier's heat; only the probe tips go into the hot air.



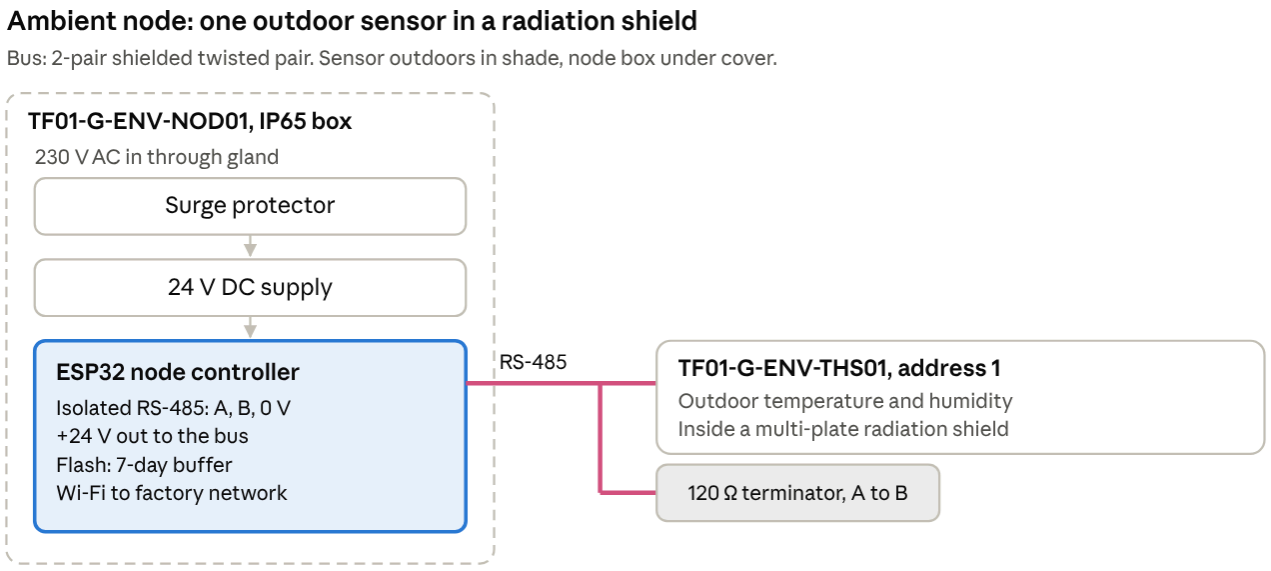
drier node · thermocouple module + current module

Order on bus	Device ID	Device	Modbus address (proposed)	Cable
1	TF01-L-DRY-TCM01	4-channel thermocouple module	41	2-pair STP
CH1 on TCM01	TF01-L-DRY-TCK01	K-type, inlet duct from the stove	None	K-type compensating cable, observe polarity
CH2 on TCM01	TF01-L-DRY-TCK02	K-type, made-tea outlet	None	K-type compensating cable
CH3 on TCM01	TF01-L-DRY-TCK03	K-type, exhaust duct	None	K-type compensating cable

Order on bus	Device ID	Device	Modbus address (proposed)	Cable
2	TF01-L-DRY-CTS01	Current module, drier fan and chain motor	21	2-pair STP; clamp lead as supplied
End	Terminator	120 Ω across A and B	None	None

Ambient node, 1 unit

One outdoor sensor. If the bus run is very short (a few metres), the terminator may not be needed; confirm on site.



ambient node · 1 Modbus device

Order on bus	Device ID	Device	Modbus address (proposed)	Cable
1	TF01-G-ENV-THS01	Outdoor temperature/humidity in radiation shield	1	2-pair STP, UV-resistant jacket outdoors
End	Terminator	120 Ω across A and B	None	None

Common parts for one box

Every bridge and node box uses the same base kit. Sizes are to be confirmed once the controller and module models are chosen.

Part	Specification	Per box
Enclosure	IP65 polycarbonate, with a window for the display on bridges	1
DIN rail	35 mm, cut to fit	1
Circuit breaker or fuse	On the 230 V supply	1
Surge protector	DIN-rail, 230 V AC	1
24 V DC supply	DIN-rail, rated for the controller plus all bus sensors	1
Controller	Industrial ESP32 (nodes) or ESP32-S3 bridge board (bridges), DIN-rail	1
Terminal blocks	For 230 V, 24 V, RS-485 A/B/0 V, relays	As needed
Earth bar	For the protective earth and the bus cable shields	1
Cable glands	IP68, sized to each cable	1 per cable
120 Ω terminator	For the far end of each RS-485 bus	1 per node
External Wi-Fi antenna	Only if the enclosure blocks the signal; confirm on site	0 or 1
ID label	Engraved or printed with the device ID	1

Totals across this doc: 7 scale bridges, 18 zone nodes, and on the buses 43 temperature/humidity sensors, 3 condensing-rated probes, 8 IR sensors, 1 thermocouple module with 3 thermocouples, and 21 current modules. These match the v2 design.